

Documentation of the Secluded Dark Matter Toolkit

Code Structure, Boltzmann Equations, and Assumptions

Auto-generated from codebase analysis

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1 Introduction

This document provides a detailed technical description of the `secluded_DM` Python toolkit, which computes the relic abundance of dark matter (DM) in *secluded dark sector* scenarios. In these models, the DM particle X annihilates primarily into unstable mediator particles Y , which subsequently decay to Standard Model (SM) final states through a small portal coupling ε . The defining feature is the parametric hierarchy

$$\alpha_X \gg \varepsilon^2, \quad (1)$$

so that the DM annihilation rate $X\bar{X} \rightarrow YY$ is set by the dark fine-structure constant α_X , while the mediator decay rate $Y \rightarrow \text{SM}$ is controlled by the much smaller portal coupling.

The code solves the coupled Boltzmann equations for the DM yield Y_X , the mediator yield Y_Y , and the dark-to-SM temperature ratio $\xi = T_h/T_{\text{SM}}$, incorporating:

- $2 \rightarrow 2$ annihilation: $X\bar{X} \leftrightarrow YY$
- $3 \rightarrow 2$ cannibal processes: $YYY \rightarrow YY$, $YYX \rightarrow YX$, $YXX \rightarrow XX$
- Mediator decay: $Y \rightarrow \text{SM}$ with portal coupling ε
- Hidden sector heating and entropy injection

Five portal models are implemented: kinetic mixing with hypercharge (`VectorPortal`), $B-L$ gauge boson mixing (`BLPortal`), L_i-L_j gauge boson mixing (`LiLjPortal`), baryon-number gauge boson mixing (`BaryonPortal`), and scalar mixing with the SM Higgs boson (`HiggsPortal`). Observational constraints from the DM relic abundance, Big Bang Nucleosynthesis (BBN) hadronic injection bounds, and direct detection experiments (particularly the combination of LZ and XENONnT bounds) are computed for each model.

2 Cosmological Background

2.1 SM Thermal Bath and Hubble Rate

The SM thermal background is described by the relativistic degrees of freedom $g_*(T)$ and $g_{*S}(T)$, loaded from tabulated data files (covering $T \sim 0.1$ MeV to ~ 100 GeV) and interpolated linearly in T . The key thermodynamic quantities are:

$$\rho_{\text{rad}}(T) = \frac{\pi^2}{30} g_*(T) T^4 \quad (\text{radiation energy density}) \quad (2)$$

$$s(T) = \frac{2\pi^2}{45} g_{*S}(T) T^3 \quad (\text{entropy density}) \quad (3)$$

$$H(T) = \sqrt{\frac{\pi^2}{90} g_*(T)} \frac{T^2}{M_{\text{Pl}}} \quad (\text{Hubble rate, SM only}) \quad (4)$$

where $M_{\text{Pl}} = 2.435 \times 10^{18}$ GeV is the reduced Planck mass. The factor

$$\tilde{g}(T) = 1 + \frac{1}{3} \frac{d \ln g_{*S}}{d \ln T} \quad (5)$$

accounts for the temperature dependence of g_{*S} in the entropy conservation equation ($\tilde{g} = 1$ when g_{*S} is constant).

When the hidden sector contributes non-negligibly to the total energy density, the Hubble rate is modified:

$$H_{\text{tot}} = \sqrt{H^2 + \frac{\rho_h}{3M_{\text{Pl}}^2}}, \quad \rho_h = \rho_X + \rho_Y. \quad (6)$$

2.2 Equilibrium Distributions and Energy Moments

All equilibrium quantities use the Maxwell-Boltzmann approximation (see Sec. 9 for a discussion of its validity). For a species with mass m , internal degrees of freedom g , at a temperature T , the number density and its derivative are given by:

$$n_{\text{MB}}^{\text{eq}}(z) = \frac{g m^3}{2\pi^2} \frac{K_2(z)}{z}, \quad \frac{dn^{\text{eq}}}{dT} = \frac{g m^2}{2\pi^2} \left[z K_1(z) + 3 K_2(z) \right] \quad (7)$$

where $z \equiv m/T$ and $K_\nu(z)$ is the modified Bessel functions of the second kind. A useful combination is the logarithmic derivative

$$\lambda(z) \equiv \frac{d \ln n^{\text{eq}}}{d \ln T} = z \frac{K_1(z)}{K_2(z)} + 3, \quad (8)$$

which appears in the Boltzmann equations whenever equilibrium tracking is needed.

Higher moments of the distribution give the energy density and pressure,

$$\rho = n^{\text{eq}} B_1, \quad \rho + P = n^{\text{eq}} B_2, \quad (9)$$

where

$$B_1(z, m) = m \left(\frac{K_1(z)}{K_2(z)} + \frac{3}{z} \right), \quad B_2(z, m) = m \left(\frac{K_1(z)}{K_2(z)} + \frac{4}{z} \right) \quad (10)$$

which can be thought of as the average energy per particle, $B_1 = \langle E \rangle/n$, and the enthalpy per particle, $B_2 = (\rho + P)/n$, respectively. Note that the difference $B_2 - B_1 = m/z = T$ recovers the ideal-gas relation $P = nT$. Their temperature derivatives are

$$\frac{dB_1}{dT} = 3 - z^2 \left[R^2 - 1 + \frac{3R}{z} \right], \quad (11)$$

$$\frac{dB_2}{dT} = 4 - z^2 \left[R^2 - 1 + \frac{3R}{z} \right], \quad (12)$$

where $R \equiv K_1(z)/K_2(z)$.

Another useful quantity that is implemented in the Boltzmann equations is the logarithmic equilibrium yield, $\ln Y_i^{\text{eq}} = \ln n^{\text{eq}} - \ln s$, with $\ln 2$ added if antiparticles are included.

Code implementation: asymptotic expansions. In our code, the density is set to zero for $z > 500$ to avoid underflow and is doubled when antiparticles are present. For $z \geq 300$, the equilibrium number density is computed via the asymptotic form

$$\ln n^{\text{eq}} = \ln \left(\frac{g m^3}{2\pi^2} \right) + \frac{1}{2} \ln \frac{\pi}{2z} - z + \ln \left(1 + \frac{15}{8z} + \frac{105}{128z^2} \right) - \ln z. \quad (13)$$

The Bessel ratio $R = K_1(z)/K_2(z)$, which enters B_1 , B_2 , λ , and their derivatives, is computed directly from `scipy.special.kv` for $z < 50$ while for $z \geq 50$ the following expansion is used instead:

$$\frac{K_1(z)}{K_2(z)} \approx 1 - \frac{3}{2z} + \frac{15}{8z^2} - \frac{105}{16z^3}. \quad (14)$$

In the same regime, the bracket in the B -factor derivatives simplifies to $3/(2z^2) - 15/(4z^3)$.

Full quantum statistics (species_thermo): A separate method computes (ρ, P, n) with full Fermi-Dirac or Bose-Einstein integrals using a three-regime approach (ultra-relativistic analytic for $z \leq 10^{-3}$, numerical integral for $10^{-3} < z < 1$, MB for $z \geq 1$). This is available for auxiliary calculations but is *not* used in the Boltzmann solver itself.

3 Dark Sector Models

All models inherit from the abstract base class `DarkSectorModel`, which defines the interface:

- Masses: m_X, m_Y (GeV)
- Internal DOF: g_X, g_Y
- Whether to include antiparticles for X and Y
- Initial dark-to-SM temperature ratio: ξ_{ini}
- Cross sections: $\langle\sigma v\rangle_{X\bar{X}\rightarrow YY}, \langle\sigma v^2\rangle_{3\rightarrow 2}$
- Decay width: $\Gamma(Y \rightarrow \text{SM})$
- Nucleon Direct detection: σ_{SI}

3.1 VectorPortal: Kinetic Mixing with Hypercharge

The DM particle X is a Dirac fermion ($g_X = 2$, includes antiparticles) charged under a dark $U(1)_D$ gauge symmetry. The mediator $Y \equiv Z'$ is the associated dark photon ($g_Y = 3$, massive spin-1, no antiparticle doubling). The dark photon mixes kinetically with the SM hypercharge gauge boson via the mixing parameter ε . We first describe the dark-sector processes — $2 \rightarrow 2$ annihilations and $3 \rightarrow 2$ cannibal reactions — and then the connection to the visible sector through kinetic mixing, which controls both the mediator decay width and the DM–nucleon scattering cross section.

3.1.1 Dark-sector processes

The thermally averaged $2 \rightarrow 2$ annihilation cross section for $X\bar{X} \rightarrow Z'Z'$ is

$$\langle\sigma v\rangle_{X\bar{X}\rightarrow Z'Z'} = 2\left(a + \frac{6b}{x_X}\right), \quad (15)$$

where $x_X = m_X/T_h$, $r_v \equiv m_Y/m_X$, and

$$a = \frac{2\pi\alpha_X^2}{m_X^2} \frac{(1 - r_v^2)^{3/2}}{(2 - r_v^2)^2}, \quad (16)$$

$$b = \frac{\pi\alpha_X^2}{12m_X^2} \frac{\sqrt{1 - r_v^2}(24 + 28r_v^2 - 36r_v^4 + 17r_v^6)}{(2 - r_v^2)^4} \quad (17)$$

are respectively the s -wave and p -wave coefficients. For the background evolution solver, where the p -wave contribution is subdominant at late times ($x \gg 1$), only the s -wave piece $\langle\sigma v\rangle_{\text{s-wave}} = 2a$ is retained.

Three $3 \rightarrow 2$ cannibal processes contribute to number-changing interactions within the dark sector. The dominant channel for the freeze-out of cannibal processes is $YYX \rightarrow YX$, whose thermally averaged cross section

$$\langle\sigma v^2\rangle_{YYX\rightarrow YX} = \Delta_3 \frac{\alpha_X^3}{m_X^5} \quad (18)$$

is computed exactly at tree level, with the coefficient

$$\Delta_3 = \frac{4\pi^2 r^5 \sqrt{\frac{4}{3}r(r+2) + 1} \mathcal{N}_3(r)}{(r+2)^3 (2r+1)^4 [2r^2(r+2) + r - 1]^2} \quad (19)$$

where $r = m_X/m_Y$ and

$$\mathcal{N}_3(r) = 2r \left(r \left(r \left(2r \left(r \left(8r(10r+71) + 1683 \right) + 2798 \right) + 6107 \right) + 4722 \right) + 2335 \right) + 578 \right) + 195. \quad (20)$$

The remaining two channels are [1]:

$$\langle \sigma v^2 \rangle_{YYY \rightarrow YY} = \Delta_1 \frac{\alpha_X^5 T_h^7}{m_X^{12}}, \quad (21)$$

$$\langle \sigma v^2 \rangle_{YXX \rightarrow XX} = \Delta_2 \frac{\alpha_X^3}{m_X^5}, \quad (22)$$

where Δ_2 is also computed exactly at tree level:

$$\Delta_2 = \frac{\pi^2 r^3 (1 + 4r)^{3/2} \mathcal{N}_2(r)}{6 (1 + 2r)^3 (2r^2 + r - 1)^2} \quad (23)$$

with

$$\mathcal{N}_2(r) = 256r^6 - 384r^5 + 536r^4 - 96r^3 + 38r^2 + 72r + 31. \quad (24)$$

The $YYY \rightarrow YY$ rate is loop-suppressed relative to the other two channels (α_X^5 vs. α_X^3) and carries an explicit temperature dependence, making it negligible for the evolution of the dark sector. We retain it for completeness with a phenomenological coefficient $\Delta_1 = 0.5$.

3.1.2 Connection to the Standard Model

After restoring canonical kinetic terms, the Z' acquires couplings to SM fermions proportional to ε . For a fermion f with electric charge Q_f and weak hypercharges Y_L , Y_R , the left- and right-handed couplings are [1]

$$g_L = \varepsilon \frac{m_{Z'}^2 g_Y^{\text{SM}} Y_L - M_Z^2 g_W s_W c_W Q_f}{M_Z^2 - m_{Z'}^2}, \quad (25)$$

$$g_R = \varepsilon \frac{m_{Z'}^2 g_Y^{\text{SM}} Y_R - M_Z^2 g_W s_W c_W Q_f}{M_Z^2 - m_{Z'}^2}, \quad (26)$$

where g_W and g_Y^{SM} are the $SU(2)_L$ and $U(1)_Y$ gauge couplings, $s_W \equiv \sin \theta_W$ and $c_W \equiv \cos \theta_W$ with θ_W the Weinberg angle, and M_Z is the Z -boson mass. The vector and axial couplings to each fermion species are then

$$g_{fV} = \frac{g_R + g_L}{2}, \quad g_{fA} = \frac{g_R - g_L}{2}. \quad (27)$$

Mediator decay width. These couplings determine the Z' decay width. The partial width into a fermion pair is

$$\Gamma(Z' \rightarrow f\bar{f}) = \frac{N_c^f m_{Z'}}{12\pi} \sqrt{1 - r_f^2} \left[g_{fV}^2 (1 + \tfrac{1}{2} r_f^2) + g_{fA}^2 (1 - r_f^2) \right] \quad (28)$$

where $r_f = 2m_f/m_{Z'}$ and N_c^f is the colour factor (3 for quarks, 1 for leptons). The total width is the sum over all kinematically accessible channels ($m_{Z'} > 2m_f$):

$$\Gamma_{Z'}^{\text{tot}} = \sum_{f: m_{Z'} > 2m_f} \Gamma(Z' \rightarrow f\bar{f}). \quad (29)$$

Direct-detection cross section. The same kinetic mixing generates a spin-independent DM–nucleon scattering cross section through Z – Z' exchange, following [2]. The Z – Z' mixing angle η is defined by

$$\tan(2\eta) = \frac{-2\varepsilon s_W M_Z^2}{M_Z^2 - m_{Z'}^2}, \quad (30)$$

and the DM couplings to the mass eigenstates are $V_X^Z = g_D \sin \eta$ and $V_X^{Z'} = g_D \cos \eta$, with $g_D = \sqrt{4\pi\alpha_X}$. The quark couplings receive contributions from both the electromagnetic and weak neutral currents:

$$V_q^Z = \sqrt{4\pi\alpha_{\text{EM}}} \varepsilon \sin \eta Q_q + \frac{\sqrt{4\pi\alpha_{\text{EM}}}}{s_W c_W} (\cos \eta + \varepsilon s_W \sin \eta) (T_3^q - s_W^2 Q_q), \quad (31)$$

$$V_q^{Z'} = \sqrt{4\pi\alpha_{\text{EM}}} \varepsilon \cos \eta Q_q + \frac{\sqrt{4\pi\alpha_{\text{EM}}}}{s_W c_W} (\sin \eta - \varepsilon s_W \cos \eta) (T_3^q - s_W^2 Q_q), \quad (32)$$

where $\alpha_{\text{EM}} \simeq 1/137$ is the electromagnetic fine-structure constant, Q_q the electric charge of quark q , and T_3^q its weak isospin. The spin-independent cross section per nucleon is then

$$\sigma_{\text{SI}} = \frac{\mu^2}{\pi} \left[\sum_{i \in \{Z, Z'\}} \frac{V_X^i [\mathcal{Z}(2V_u^i + V_d^i) + (\mathcal{A} - \mathcal{Z})(V_u^i + 2V_d^i)]}{\mathcal{A} m_i^2} \right]^2 \times \hbar^2 c^2 \quad (33)$$

where $\mu = m_X m_N / (m_X + m_N)$ is the reduced mass with $m_N = (m_p + m_n)/2$ the average nucleon mass, and $\mathcal{Z}$ and $\mathcal{A}$ are the atomic and mass numbers of the target nucleus.

Two notable features appear in the cross section as a function of $m_{Z'}$. First, the couplings in Eqs. (25)–(26) exhibit a resonance at $m_{Z'} = M_Z$, where σ_{SI} is strongly enhanced and the required ε is correspondingly minimised. In practice this pole would be regularised by the finite width $\Gamma_{Z'}$, which we neglect here; nevertheless, the enhancement remains significant in the vicinity of the Z pole. Second, the coherent sum over Z and Z' exchange leads to an exact cancellation between the two amplitudes above the pole. For $m_{Z'} > M_Z$ the mixing angle η changes sign, flipping the relative sign of the two amplitudes. At a critical mass

$$m_{Z'}^c \simeq 129.4 \text{ GeV} \quad (\approx 1.42 M_Z) \quad (\text{Xe-131 target}), \quad (34)$$

the two amplitudes cancel exactly and $\sigma_{\text{SI}} = 0$, independent of ε and α_X to leading order. At this blind spot the required $\varepsilon \rightarrow \infty$, making the model undetectable. The position of the zero depends on the target nucleus through the $(\mathcal{Z}, \mathcal{A})$ -weighted quark couplings but is independent of the dark-sector parameters.

The default parameter values used in the `VectorPortal` class are summarised in Table 1.

Parameter	Symbol	Default
DM mass	m_X	100 GeV
Mediator mass	m_Y	30 GeV
Dark coupling	α_X	10^{-2}
DM DOF	g_X	2 (Dirac fermion)
Mediator DOF	g_Y	3 (massive vector)
Cannibal rescaling	Δ_1	0.5
Cannibal rescaling	Δ_2, Δ_3	exact tree-level
Initial temperature ratio	ξ_{ini}	1.0

Table 1: Default parameters for the `VectorPortal` model.

3.2 BLPortal: $B - L$ Kinetic Mixing

The hidden-sector Z' mixes kinetically with the $U(1)_{B-L}$ gauge boson Z_{B-L} . The coupling to SM fermion f is purely vectorial [3]:

$$g_{fV} = \varepsilon g_{B-L} (B - L)_f \left| \frac{m_{Z_{B-L}}^2 + m_{Z'}^2}{m_{Z_{B-L}}^2 - m_{Z'}^2} \right|, \quad g_{fA} = 0 \quad (35)$$

where g_{B-L} is the $B-L$ gauge coupling and $(B-L)_f$ the fermion charge: $+1/3$ for quarks and -1 for leptons. Since the dark sector is identical to the **VectorPortal**, all $2 \rightarrow 2$ and $3 \rightarrow 2$ cross sections (Eqs. (16)–(18)) carry over unchanged. The Z' decay width retains the same functional form as Eq. (28), but with purely vectorial couplings ($g_{fA} = 0$) and all kinematically accessible fermions contributing. For direct detection, all quarks carry the same $(B-L) = 1/3$ charge, so the coupling is isospin-universal: proton and neutron receive equal contributions. Evaluating g_{qV} from Eq. (35) with $(B-L)_q = 1/3$, the three valence quarks in a nucleon contribute coherently, giving

$$\sigma_{\text{SI}} = \frac{\mu^2}{\pi} \left(\frac{3 g_D g_{qV}}{m_{Z'}^2} \right)^2 \times \hbar^2 c^2 \quad (36)$$

where g_D and μ are as defined in Sec. 3.1.2. Unlike the hypercharge portal, only a single mediator contributes and no blind spot arises.

3.3 BaryonPortal: Baryon Number Kinetic Mixing

The Z' mixes kinetically with the gauge boson of $U(1)_B$, acquiring purely vectorial couplings to quarks proportional to their baryon number [3]:

$$g_{fV} = \varepsilon g_B B_f \left| \frac{m_{Z_B}^2 + m_{Z'}^2}{m_{Z_B}^2 - m_{Z'}^2} \right|, \quad g_{fA} = 0 \quad (37)$$

where g_B is the $U(1)_B$ gauge coupling, m_{Z_B} the associated gauge boson mass, and $B_f = 1/3$ for all quarks. Leptons are uncharged, so the Z' decays exclusively to hadronic channels. The dark-sector cross sections are again identical to the **VectorPortal**. Since all quarks carry the same baryon charge, the direct-detection cross section is isospin-universal and takes the same form as Eq. (36), with g_{qV} evaluated from Eq. (37).

3.4 LiLjPortal: $L_i - L_j$ Kinetic Mixing

The Z' mixes kinetically with the gauge boson of $U(1)_{L_i-L_j}$, acquiring purely vectorial couplings to leptons and their neutrinos of the two involved families [3]: flavour i carries charge $+1$ and flavour j carries charge -1 , while quarks are uncharged. Three flavour combinations are supported: $e-\mu$, $\mu-\tau$, and $e-\tau$. The dark-sector cross sections are identical to those of the **VectorPortal** (Eqs. (16)–(18)). Because quarks are uncharged, the Z' decays exclusively to leptons at tree level and the spin-independent DM–nucleon cross section vanishes, $\sigma_{\text{SI}} = 0$. A nonzero σ_{SI} arises at loop level, but the resulting cross section is too small to produce meaningful direct-detection constraints unless the couplings are very large, so we neglect it.

3.5 HiggsPortal: Scalar Mixing

The DM particle X is a Majorana fermion ($g_X = 2$, no antiparticle doubling) coupled to a real scalar mediator ϕ ($g_Y = 1$) that mixes with the SM Higgs boson through the mixing angle ε , following the conventions of [3]. In the small-angle limit $\sin \varepsilon \approx \varepsilon$. Two couplings parametrise the dark-sector Yukawa interactions:

$$\mathcal{L} \supset -\frac{\lambda_s}{2} \phi \bar{X} X - \frac{i\lambda_p}{2} \phi \bar{X} \gamma_5 X \quad (38)$$

where λ_s and λ_p are the scalar and pseudoscalar couplings respectively. We first describe the dark-sector processes and then the connection to the SM through scalar mixing, which controls both the mediator decay width and the DM–nucleon scattering cross section.

3.5.1 Dark-sector processes

The thermally averaged $2 \rightarrow 2$ annihilation cross section for $X\bar{X} \rightarrow \phi\phi$ is

$$\langle\sigma v\rangle_{X\bar{X}\rightarrow\phi\phi} = 2\left(a + \frac{6b}{x_X}\right), \quad (39)$$

where $x_X = m_X/T_h$, $r \equiv m_Y/m_X$, and

$$a = \frac{2\sqrt{1-r^2}\lambda_p^2\lambda_s^2}{m_X^2\pi(r^2-2)^2}, \quad (40)$$

$$b = \frac{1}{12m_X^2\pi\sqrt{1-r^2}(r^2-2)^4} \left[-2(r^2-1)^3\lambda_p^4 + 3(r^6-8r^4+20r^2-12)\lambda_s^2\lambda_p^2 + 2(-2r^6+10r^4-17r^2+9)\lambda_s^4 \right] \quad (41)$$

are respectively the s -wave and p -wave coefficients. Note that the s -wave vanishes when either $\lambda_s = 0$ or $\lambda_p = 0$, so both couplings must be nonzero for a nonvanishing leading-order cross section. For the background evolution solver, only the s -wave piece $\langle\sigma v\rangle_{s\text{-wave}} = 2a$ is retained, since at late times ($x \gg x_{\text{FO}}$) the p -wave contribution is Boltzmann-suppressed and negligible. For $x_X < 1$ the p -wave term is also dropped, which is inconsequential since the dark sector is still in full equilibrium in this regime.

Three $3 \rightarrow 2$ cannibal processes are included, all computed exactly at tree level. In analogy to the **VectorPortal** (Eq. (18)), the dominant channel for cannibal freeze-out is $\phi\phi X \rightarrow \phi X$, whose cross section can be written as

$$\langle\sigma v^2\rangle_{\phi\phi X\rightarrow\phi X} = \Delta_3(r, \rho) \frac{\lambda_s^3\lambda_p^3}{m_X^5} \quad (42)$$

where $r \equiv m_X/m_Y$, $\rho \equiv \lambda_p/\lambda_s$, and

$$\begin{aligned} \Delta_3(r, \rho) = & \frac{9\sqrt{3}r^5\sqrt{3+8r+4r^2}}{64\pi(r+2)^3(2r+1)^4(2r^3+4r^2+r-1)^2} \times \left[(1+2r)^5(3+2r) \right. \\ & + (9+16r+180r^2+400r^3+288r^4+64r^5)(1+2r)^2\rho^2 + 3(-1+2r+2r^2)^2\rho^6 \\ & \left. + (9+8r+140r^2+888r^3+2220r^4+3360r^5+3024r^6+1408r^7+256r^8)\rho^4 \right]. \quad (43) \end{aligned}$$

The second channel, $\phi XX \rightarrow XX$, shares the same factorisation:

$$\langle\sigma v^2\rangle_{\phi XX\rightarrow XX} = \Delta_2(r, \rho) \frac{\lambda_s^3\lambda_p^3}{m_X^5} \quad (44)$$

with

$$\begin{aligned} \Delta_2(r, \rho) = & \frac{r^3\sqrt{1+4r}}{64\pi(1+2r)^3(-1+r+2r^2)^2} \times \left[(1+8r)(1+\rho^2)^3 + 4r^2(2+21\rho^2+20\rho^4+\rho^6) \right. \\ & - 16r^3(4-7\rho^2-6\rho^4+5\rho^6) - 16r^4(7-6\rho^2-25\rho^4+4\rho^6) \\ & \left. + 128r^5(1+9\rho^2+14\rho^4+2\rho^6) + 256r^6(1+2\rho^2)^2 \right]. \quad (45) \end{aligned}$$

This channel is generally subdominant but is included for completeness. The third channel, $\phi\phi\phi \rightarrow \phi\phi$, factorises with a different coupling dependence:

$$\langle\sigma v^2\rangle_{\phi\phi\phi\rightarrow\phi\phi} = \Delta_1(r, \rho) \frac{\lambda_s^5\lambda_p^5}{m_X^5} \quad (46)$$

with

$$\Delta_1(r, \rho) = \frac{\sqrt{5}}{3\pi^5} \frac{r^3 (3 + 10\rho^2 + 15\rho^4)^2}{\rho^3}. \quad (47)$$

Unlike the tree-level channels above, $\phi\phi\phi \rightarrow \phi\phi$ proceeds through a one-loop pentagon diagram (a closed X loop with five external ϕ legs) and the expression above is derived in the heavy-fermion limit $r \gg 1$, where all external momenta are negligible compared to the loop momentum. This is self-consistent as the $\phi\phi\phi \rightarrow \phi\phi$ channel can only become competitive with $\phi\phi X \rightarrow \phi X$ precisely in this regime, since at low r the dark matter X is comparably abundant to ϕ and the tree-level process wins without any loop penalty. At large r , however, X is exponentially Boltzmann-suppressed in the initial state, and the cannibal process — which only requires X as a virtual particle inside the loop — sidesteps this suppression, compensating the $\sim \lambda^4/(16\pi^2)^2$ loop cost.

3.5.2 Connection to the Standard Model

Mediator decay width. The mediator ϕ decays to SM particles through its mixing with the Higgs boson. All partial widths are proportional to $\sin^2\epsilon \approx \epsilon^2$, and we define the normalised total width $\Gamma_{\text{norm}}(m_\phi) \equiv \Gamma_{\text{tot}}/\sin^2\epsilon$ so that the physical width is

$$\Gamma(\phi \rightarrow \text{SM}) = \epsilon^2 \Gamma_{\text{norm}}(m_\phi). \quad (48)$$

The SM-like contribution to Γ_{norm} is computed using a hybrid approach across four mass regimes:

- $m_\phi < 5 \text{ GeV}$: the `scalar_portal` package [4], which uses a dispersive analysis below 2 GeV (leptonic decays, hadronic channels via form factors, multi-meson final states) and perturbative QCD with running quark masses ($N_f = 4$) above 2 GeV.
- $5 \text{ GeV} \leq m_\phi \leq 500 \text{ GeV}$: the `HDECAY` Fortran program [5, 6], which includes higher-order QCD corrections, off-shell decays to WW^*/ZZ^* , and loop-induced channels ($gg, \gamma\gamma$).
- $m_\phi > 500 \text{ GeV}$: tree-level decay widths with exact kinematics, since `HDECAY` is not reliable in this regime due to un-resummed Sudakov logarithms.

If $m_\phi > 2m_h$, an additional contribution from $\phi \rightarrow hh$ arises in the singlet extension:

$$\frac{\Gamma(\phi \rightarrow hh)}{\sin^2\epsilon} = \frac{(2m_\phi^2 + m_h^2)^2}{128\pi m_\phi v_H^2} \sqrt{1 - \frac{4m_h^2}{m_\phi^2}} \quad (49)$$

with $v_H = 246.22 \text{ GeV}$ the Higgs vacuum expectation value. This channel can be switched off by setting `hh_coupling='zero'`.

Direct-detection cross section. The spin-independent DM–nucleon scattering cross section proceeds via t -channel exchange of the two mixed scalars h and ϕ . After electroweak symmetry breaking, the mass mixing $\phi \rightarrow \phi - \epsilon h$ generates an effective ϕ –nucleon coupling proportional to $\epsilon \lambda_s$. In the small-mixing limit, the h - and ϕ -exchange amplitudes add coherently with opposite sign (since the mixing rotates them in opposite directions), giving a per-nucleon amplitude proportional to the propagator difference $1/m_Y^2 - 1/m_h^2$. The resulting cross section is

$$\sigma_{\text{SI}} = \frac{4\mu^2}{\pi} a_N^2 \hbar^2 c^2, \quad a_N = \frac{\lambda_s \epsilon f_N m_N}{v_H} \left(\frac{1}{m_Y^2} - \frac{1}{m_h^2} \right) \quad (50)$$

where $f_N \simeq 0.30$ is the Higgs–nucleon form factor, m_h the SM Higgs mass, and μ the DM–nucleon reduced mass. The factor of 4 reflects the Majorana nature of X (identical-particle combinatorics). This result is isospin-universal and independent of the target nucleus composition, since the coupling to nucleons proceeds entirely through the Higgs–nucleon vertex. The numerical estimate in [1] (their Eq. 67) is consistent with this expression in the limit $m_Y \ll m_h$. The propagator difference vanishes at $m_Y = m_h$, where the two amplitudes cancel exactly and $\sigma_{\text{SI}} = 0$.

At this blind spot the required $\varepsilon \rightarrow \infty$, making the model undetectable by direct-detection experiments regardless of the dark-sector couplings.

The default parameter values used in the `HiggsPortal` class are summarised in Table 2.

Parameter	Symbol	Default
DM mass	m_X	100 GeV
Mediator mass	m_Y	30 GeV
Dark couplings	$\lambda_s = \lambda_p$	10^{-2}
DM DOF	g_X	2 (Majorana fermion)
Mediator DOF	g_Y	1 (real scalar)
Cannibal rescaling	$\Delta_1, \Delta_2, \Delta_3$	exact tree-level
Initial temperature ratio	ξ_{ini}	1.0
$\phi \rightarrow hh$ coupling	<code>hh_coupling</code>	'singlet'

Table 2: Default parameters for the `HiggsPortal` model.

4 Boltzmann Equations

The dark sector evolves as a coupled system of three ODEs in the independent variable $x = m_X/T_{\text{SM}}$, tracking the dark-to-SM temperature ratio $\ln \xi$ (with $\xi = T_h/T_{\text{SM}}$) and the chemical potential shifts $\bar{\mu}_X, \bar{\mu}_Y$. The yields are parametrized as

$$Y_i = Y_i^{\text{eq}} e^{\bar{\mu}_i}, \quad i \in \{X, Y\}, \quad (51)$$

which avoids numerical problems in the ultra-relativistic regime ($Y_i \approx Y_i^{\text{eq}}$, i.e. $\bar{\mu}_i \approx 0$) and naturally captures the exponential departure from equilibrium during freeze-out.

4.1 Full System

The complete system solved during the out-of-equilibrium phase (Phase 2 of the QSSA solver, Sec. 5.1) is:

Dark-sector temperature (energy conservation):

$$\frac{d \ln \xi}{dx} = \frac{E + FA + GC}{1 - FB - GD} \quad (52)$$

DM chemical potential ($X\bar{X} \leftrightarrow YY$ annihilation):

$$\frac{d\bar{\mu}_X}{dx} = A + B \frac{d \ln \xi}{dx} \quad (53)$$

Mediator chemical potential (annihilation source + cannibal processes):

$$\frac{d\bar{\mu}_Y}{dx} = C + D \frac{d \ln \xi}{dx} \quad (54)$$

The six coefficients A – G entering these equations are defined as follows.

DM coefficients:

$$A = -\frac{s\tilde{g}}{H_{\text{tot}} x} \frac{\langle\sigma v\rangle_{X\bar{X}\rightarrow YY}}{2} \left(Y_X - \frac{(Y_X^{\text{eq}})^2 Y_Y^2}{Y_X (Y_Y^{\text{eq}})^2} \right) + \frac{\lambda_X - 3\tilde{g}}{x} \quad (55)$$

$$B = -\lambda_X \quad (56)$$

The first term in A is the collision integral for $X\bar{X} \leftrightarrow YY$; the second accounts for dilution from SM entropy production.

Mediator coefficients:

$$C = \frac{s\tilde{g}}{H_{\text{tot}} x} \frac{\langle\sigma v\rangle}{2} \frac{Y_X^2 - (Y_X^{\text{eq}})^2 (Y_Y/Y_Y^{\text{eq}})^2}{Y_Y} - \Gamma_{\text{can}} (1 - e^{-\bar{\mu}_Y}) + \frac{\lambda_Y - 3\tilde{g}}{x} \quad (57)$$

$$D = -\lambda_Y \quad (58)$$

where the dimensionless cannibal rate (in units of the Hubble rate) collecting all number-changing $3 \rightarrow 2$ processes is:

$$\frac{\Gamma_{\text{can}}}{H_{\text{tot}}} = \frac{s^2 \tilde{g}}{H_{\text{tot}} x} \left[Y_Y^2 \langle\sigma v^2\rangle_{YY\rightarrow YY} + Y_Y Y_X \langle\sigma v^2\rangle_{YYX\rightarrow YX} + Y_X^2 \langle\sigma v^2\rangle_{YXX\rightarrow XX} \right] \quad (59)$$

Temperature-equation coefficients (from dark-sector energy conservation):

$$E = \frac{1}{x} \left(1 - 3\tilde{g} \frac{n_X B_2^X + n_Y B_2^Y}{\tilde{\mathcal{D}}} \right) \quad (60)$$

$$F = -\frac{B_1^X n_X}{\tilde{\mathcal{D}}}, \quad G = -\frac{B_1^Y n_Y}{\tilde{\mathcal{D}}} \quad (61)$$

with the energy denominator

$$\tilde{\mathcal{D}} = T_h \left(n_X \frac{dB_1^X}{dT_h} + n_Y \frac{dB_1^Y}{dT_h} \right) + B_1^X n_X \lambda_X + B_1^Y n_Y \lambda_Y. \quad (62)$$

The B -factors and λ_i are defined in Sec. 2.2.

5 Solver Implementations

The primary solver is the three-phase QSSA method described below. Three additional legacy solvers are retained in the codebase for testing and cross-checks but are not used in production (see Sec. 5.2).

5.1 Three-Phase QSSA Solver (solve_boltzmann_chempot_QSSA)

This is the primary (and default) solver. It exploits the physical hierarchy of time-scales in the secluded dark sector: annihilation freezes out first while the mediator is still held in equilibrium by cannibal processes, and only later does the mediator itself depart from equilibrium. Each phase solves the appropriate reduction of the full system (Sec. 4.1).

Throughout this section, the dimensionless ratios $\Gamma_{\text{ann}}/H_{\text{tot}}$ and $\Gamma_{\text{can}}/H_{\text{tot}}$ (Eqs. (64), (59)) measure the annihilation and cannibal interaction rates relative to the Hubble rate. A ratio $\gg 1$ means the process is fast enough to maintain equilibrium; a ratio $\lesssim 1$ means the process is freezing out.

Phase 1 (Equilibrium): $\bar{\mu}_X = \bar{\mu}_Y = 0$. Both species track their thermal equilibrium abundances and only the dark-sector temperature $\ln \xi$ evolves:

$$\left. \frac{d \ln \xi}{dx} \right|_{\text{eq}} = \frac{1}{x} \left(1 - \left(3 + \frac{d \ln g_{*S}}{d \ln T} \right) \frac{n_Y^{\text{eq}} B_2^Y + n_X^{\text{eq}} B_2^X}{T_h \left(B_1^X \frac{dn_X^{\text{eq}}}{dT_h} + \frac{dB_1^X}{dT_h} n_X^{\text{eq}} + B_1^Y \frac{dn_Y^{\text{eq}}}{dT_h} + \frac{dB_1^Y}{dT_h} n_Y^{\text{eq}} \right)} \right) \quad (63)$$

The solver exits this phase when the dimensionless annihilation rate

$$\frac{\Gamma_{\text{ann}}}{H_{\text{tot}}} = \frac{s}{H_{\text{tot}}} \frac{\langle \sigma v \rangle_{X\bar{X} \rightarrow YY}}{2} Y_X \quad (64)$$

drops below Γ_{QSSA}/H (default 5×10^9), signalling the onset of DM freeze-out.

Phase 1.5 (QSSA – Y in equilibrium): $\bar{\mu}_Y = 0$, evolves $(\ln \xi, \bar{\mu}_X)$. The DM equation simplifies using the collision rate with equilibrium Y :

$$A_{\text{QSSA}} = -\frac{s \tilde{g}}{H_{\text{tot}} x} \langle \sigma v \rangle Y_X^{\text{eq}} \sinh(\bar{\mu}_X) + \frac{\lambda_X - 3\tilde{g}}{x} \quad (65)$$

The temperature equation becomes:

$$\left. \frac{d \ln \xi}{dx} \right|_{\text{QSSA}} = \frac{C_\xi + D_\xi A}{1 - B D_\xi} \quad (66)$$

where C_ξ and D_ξ are the temperature-evolution coefficients from the energy conservation equation with Y fixed at equilibrium.

The solver exits this phase when $\Gamma_{\text{can}}/H_{\text{tot}}$ (Eq. (59)) drops below `cannibal_switch_full` (default = 1), i.e. when the cannibal interaction rate falls below the Hubble rate and can no longer maintain the mediator in chemical equilibrium. At this point the full three-variable system (Sec. 4.1) is engaged.

Phase 2 (Full system): All three variables $(\ln \xi, \bar{\mu}_X, \bar{\mu}_Y)$ evolve via Eqs. (52)–(54).

x_{max} scaling: For large mass ratios, the mediator freezes out much later than the DM. The solver automatically scales the integration range:

$$x_{\text{max}} = \max(10^5, r^2 \times 10^4), \quad r = m_X/m_Y \quad (67)$$

5.2 Legacy Solvers

The following three solvers are superseded by the QSSA solver and are retained only for testing and cross-checks. They are not called by any production script or notebook.

5.2.1 Two-Phase Solver (`solve_boltzmann_chempot`)

Phase 1 (Equilibrium): $\bar{\mu}_X = \bar{\mu}_Y = 0$. Evolves $\ln \xi$ via Eq. (63). Transitions to Phase 2 when $\Gamma_{\text{ann}} < \Gamma_{\text{switch}}$ (default 10^7).

Phase 2 (Full system): All three variables $(\ln \xi, \bar{\mu}_X, \bar{\mu}_Y)$ evolve via Eqs. (52)–(54).

Limitation: Jumping directly from full equilibrium to the full three-variable system introduces stiffness near the annihilation freeze-out transition, where Y is still in equilibrium but X is departing. The QSSA solver resolves this by inserting the intermediate Phase 1.5.

5.2.2 Hybrid Solver (solve_boltzmann_hybrid)

Phase 1.5 uses Y -variables (physical yields Y_X, Y_Y) instead of chemical potentials:

$$\frac{dY_X}{dx} = -\frac{s\tilde{g}}{H_{\text{tot}}x} \frac{\langle\sigma v\rangle}{2} \left(Y_X + \frac{Y_X^{\text{eq}}}{Y_Y^{\text{eq}}} Y_Y \right) \left(Y_X - \frac{Y_X^{\text{eq}}}{Y_Y^{\text{eq}}} Y_Y \right) \quad (68)$$

$$\frac{dY_Y}{dx} = -\frac{dY_X}{dx} - \frac{s^2\tilde{g}}{H_{\text{tot}}x} (Y_Y - Y_Y^{\text{eq}}) \mathcal{S}_{3\rightarrow 2} \quad (69)$$

Transitions to Phase 2 (chemical potential variables) when either $\Gamma_{\text{can}} < \text{cannibal_switch_mu}$ or the fractional departure $|Y_X - Y_X^{\text{eq}}|/Y_X^{\text{eq}} > \text{delta_switch_mu}$.

Limitation: Tracking Y_Y directly can suffer from numerical issues when it spans many orders of magnitude during the cannibal epoch.

5.2.3 Y -Equilibrium Solver (solve_boltzmann_Yeq)

The simplest solver, assuming the mediator Y always tracks its thermal equilibrium abundance. Solves only two variables: $(\ln \xi, \bar{\mu}_X)$. Equations are identical to the QSSA Phase 1.5 but are used throughout the entire evolution. Uses the s -wave-only cross section $\langle\sigma v\rangle_{s\text{-wave}}$.

Limitation: By construction, this solver cannot capture the cannibal freeze-out of Y or its subsequent departure from equilibrium, which affects the late-time temperature evolution and mediator abundance.

5.3 Numerical Details

All solvers use `scipy.integrate.solve_ivp` with the implicit Runge-Kutta method **Radau** (L-stable, suitable for stiff problems), with maximum step size 1 in x , relative tolerance 10^{-8} , and absolute tolerance 10^{-15} (Phase 2) or 10^{-12} (Phase 1). The integration proceeds in segments over a predefined grid:

$$x \in \{10^{-2}, 10^{-1}, 0.99\} \cup \{1, 1.5, 2, \dots, 10\} \cup \{10.5, 11.5, \dots, 99.5\} \cup \{200, 500, \dots, x_{\text{max}}\} \quad (70)$$

For $x < 1$, evaluation points are log-spaced; for $x \geq 1$, they are linearly spaced. Each segment uses $n_{\text{points}} = 500$ evaluation points.

Clipping and regularisation: $\ln \xi$ is clipped to $[-20, 20]$; $\bar{\mu}_i$ is clipped to $[-10/x, 10/x]$ per Hubble time; yield exponentials are clipped to $[-700, 700]$; small densities ($< 10^{-300}$) are replaced with `copysign(10-300, .)`.

Initial conditions: The initial temperature ratio at $x_0 = x_{\text{min}}$ (default 10^{-3}) is set by entropy conservation from a high reference temperature T_{∞} (default 10^{14} GeV):

$$\ln \xi(x_0) = \frac{1}{3} \ln \left(\frac{g_{*S}(m_X/x_0)}{g_{*S}(T_{\infty})} \right) + \ln \xi_{\text{ini}} \quad (71)$$

Chemical potentials start at zero: $\bar{\mu}_X(x_0) = \bar{\mu}_Y(x_0) = 0$.

Convergence: Two modes are supported, checked when $x > 2x_{\text{FO}}$ (where x_{FO} is defined as the point where $\bar{\mu}_X$ first exceeds 0.05):

- `dlnxi_NR` (default): $\left| \frac{d \ln \xi}{dx} \right|_{\text{num}} - \frac{1-2\tilde{g}}{x} \cdot x < 10^{-2}$
- `dlnYX`: $\left| \frac{d \ln Y_X}{d \ln x} \right| < \text{convergence_threshold}$

6 Background Evolution (Late-Time Y Decay)

After the Boltzmann solver determines the freeze-out yields, the `solve_background` method evolves the system through the epoch of mediator decay into SM particles, tracking entropy injection and the resulting dilution of the DM relic abundance.

The system is evolved in e -folds $N = \ln(a/a_0)$ using three log-scaled variables: $y_0 = \ln(T/T_0)$, $y_1 = \ln(\rho_Y/\rho_{Y,0})$, $y_2 = \ln(\rho_X/\rho_{X,0})$, where subscript 0 denotes values at the start of the background evolution (end of Boltzmann solver). The evolution equations are:

$$\frac{d \ln T}{dN} = -\frac{1}{\tilde{g}} \left(1 - \frac{\rho_Y \Gamma_Y}{H \cdot 3sT} \right) \quad (72)$$

$$\frac{d \ln \rho_Y}{dN} = - \left(3 + \frac{\Gamma_Y}{H} \right) \quad (73)$$

$$\frac{d \ln \rho_X}{dN} = -3 - \frac{\langle \sigma v \rangle_{s\text{-wave}} \rho_X}{2m_X H} \quad (74)$$

with the total Hubble rate $H = \sqrt{(\rho_{\text{rad}} + \rho_X + \rho_Y)/(3M_{\text{Pl}}^2)}$ and decay width $\Gamma_Y = \varepsilon^2 \Gamma_{\text{norm}}(Y \rightarrow \text{SM})$. The first equation captures entropy injection from $Y \rightarrow \text{SM}$ decay; the second combines Hubble dilution and decay; the third includes residual DM annihilation.

The comoving entropy $S = s a^3$ is tracked, and the entropy ratio $S_{\text{ratio}} = S_{\text{final}}/S_0$ quantifies the total entropy injection. For prompt Y decay ($\Gamma_Y \gg H$), $S_{\text{ratio}} \approx 1$; for late decay, $S_{\text{ratio}} > 1$ and the DM yield is diluted to $Y_X^{\text{final}} = \rho_X^{\text{final}}/(m_X s_{\text{final}})$. The evolution terminates when $\rho_Y/\rho_{\text{tot}} < 10^{-12}$ (mediator fully decayed).

Y -decay correction (`correct_Y_decay`). The Boltzmann solver assumes Y is stable (no portal coupling). For moderate values of ε , the mediator decay rate Γ_Y can become comparable to the Hubble rate during the Boltzmann evolution, meaning Y should have partially decayed before the background solver begins. When `correct_Y_decay = True`, the code corrects for this by stepping through the Boltzmann trajectory and computing the cumulative decay and entropy injection that was missed:

1. At each step along the trajectory, the elapsed proper time is $dt = dx / (x H_{\text{tot}} \tilde{g})$.
 2. The surviving fraction is tracked as $f_{\text{survive}} \rightarrow f_{\text{survive}} \times e^{-\Gamma_Y dt}$.
 3. The entropy injected at the local SM temperature is accumulated: $d(\ln S) = \Delta Y_Y \cdot m_Y / T$.
- The initial conditions are then corrected before evolving the background ODE:

$$Y_{Y,0} \rightarrow Y_{Y,0} f_{\text{survive}}, \quad T_0 \rightarrow T_0 \left(\frac{S_{\text{corr}}}{S_0} \right)^{1/3}, \quad Y_{X,0} \rightarrow Y_{X,0} / \left(\frac{S_{\text{corr}}}{S_0} \right) \quad (75)$$

This correction is negligible for $\varepsilon \lesssim 10^{-11}$ (where $\Gamma_Y \ll H$ throughout the Boltzmann evolution) but becomes important at $\varepsilon \sim 10^{-10}$ where $\Gamma_Y/H \sim \mathcal{O}(1)$ near the end of the Boltzmann phase.

7 Constraint Finding

7.1 Relic Abundance Constraint

The portal coupling ε that yields the observed DM relic abundance is found by root-finding (`brentq`) on:

$$f(\log_{10} \varepsilon) = \log_{10} Y_X^{\text{final}}(\varepsilon) - \log_{10} Y_{\text{target}} \quad (76)$$

where:

$$Y_{\text{target}} = \frac{\Omega_c h_{\text{target}}^2 \rho_c}{s_0 m_X} \quad (77)$$

with $\Omega_c h_{\text{target}}^2 = 0.12$, $\rho_c = 1.05375 \times 10^{-5} \text{ GeV } h^2 \text{ cm}^{-3}$, and $s_0 = 2891 \text{ cm}^{-3}$. Default search range: $\log_{10} \varepsilon \in [-20, -6]$.

7.2 BBN Hadronic Injection Constraint

The code uses the BBN bounds from [7] on late-decaying particles that inject hadronic energy during nucleosynthesis. The constraint is tabulated as upper bounds on $m_Y Y_Y$ [GeV] as a function of the mediator lifetime τ [s], for two injection channels ($u\bar{u}$ and $b\bar{b}$) at six reference masses (30 GeV, 100 GeV, 1 TeV, 10 TeV, 100 TeV, 1000 TeV). A 2D log-log regular grid interpolator (`RegularGridInterpolator`) is built on $(\log_{10}(m/\text{TeV}), \log_{10} \tau)$.

Outside the tabulated mass range, power-law extrapolation is used with:

$$\text{bound}(m) = \text{bound}(m_{\text{anchor}}) \times (m/m_{\text{anchor}})^\alpha, \quad (78)$$

where $\alpha_{\text{low}} = 0.50$ ($u\bar{u}$) or 0.34 ($b\bar{b}$) below 30 GeV, and $\alpha_{\text{high}} = 0.70$ for both channels above 1000 TeV. For $\tau < \tau_{\text{min}}$ (decays before BBN) no constraint is applied; for $\tau > \tau_{\text{max}}$ the bound is clamped to the longest- τ value (conservative).

For models with multiple decay channels, the effective bound is obtained by harmonic weighting:

$$\frac{1}{\text{bound}_{\text{eff}}} = \frac{\text{BR}_{u\bar{u}\text{-like}}}{\text{bound}_{u\bar{u}}} + \frac{\text{BR}_{b\bar{b}}}{\text{bound}_{b\bar{b}}} \quad (79)$$

The channel classification is: $u\bar{u}$ -like ($u\bar{u}$, $d\bar{d}$, $s\bar{s}$, $c\bar{c}$, $t\bar{t}$, gg , WW , ZZ , hh); $b\bar{b}$ -like ($b\bar{b}$, distinct fragmentation spectrum); leptonic channels (e^+e^- , $\mu^+\mu^-$, $\tau^+\tau^-$, $\nu\bar{\nu}$, $\gamma\gamma$, $Z\gamma$) carry no constraint.

The minimum portal coupling satisfying the BBN constraint is found by root-finding on $f(\log_{10} \varepsilon) = \text{bound}(m_Y, \tau(\varepsilon)) - m_Y Y_Y^{\text{frozen}}$, where the mediator lifetime is $\tau = \hbar/(\varepsilon^2 \Gamma_{\text{norm}})$.

7.3 Thermalization Floor

The minimum portal coupling for the dark sector to remain in thermal equilibrium with the SM at freeze-out is estimated as:

$$\varepsilon_{\text{therm}} = \sqrt{\frac{H(T_{\text{FO}})}{\Gamma_{\text{norm}}(Y \rightarrow \text{SM})}} \quad (80)$$

where $T_{\text{FO}} = m_X/x_{\text{FO}}$ (default $x_{\text{FO}} = 20$). This is based on the requirement $\Gamma(Y \rightarrow \text{SM}) \geq H(T_{\text{FO}})$ from Sec. 3.1.2 of arXiv:2509.08043.

7.4 Direct Detection Constraints

The portal coupling saturating a direct detection limit is found by root-finding on:

$$f(\log_{10} \varepsilon) = \log_{10} \sigma_{\text{SI}}(\varepsilon) - \log_{10} \sigma_{\text{limit}}(m_X) \quad (81)$$

Two built-in constraints are available:

- **LZ 90% CL:** Upper limit from the LZ experiment (valid for $m_X \in [9, 10000]$ GeV), loaded from tabulated data in `source/data/direct_detection_data/LZ_SI_90CL.txt`
- **Xenon neutrino floor:** Coherent neutrino-nucleus scattering background (valid for $m_X \in [0.1, 10000]$ GeV), from `NuFloors/Xe_SI.txt`

Both are interpolated log-log from tabulated $(m_X, \sigma_{\text{SI}})$ data.

Returns ∞ for models with vanishing SI cross section (e.g. `LiLjPortal`), and `NaN` for masses outside the data range.

8 Critical Mass Finder (`find_mXstar`)

The function `find_mXstar` finds the DM mass m_X^* at which the thermal freeze-out mechanism produces exactly the observed relic abundance $\Omega_c h^2 = 0.12$, for a given mass ratio $r = m_X/m_Y$. The algorithm proceeds as follows:

1. **Estimate:** An initial mass estimate is obtained from the model's s -wave cross section at unit mass:

$$\alpha_{\text{eff}} = \sqrt{\frac{\langle\sigma v\rangle_{s\text{-wave}}(m=1)}{\pi}}, \quad m_X^{\text{est}} = \alpha_{\text{eff}} \sqrt{\pi m_Y^{\text{relic}} M_{\text{Pl}}} \quad (82)$$

where $m_Y^{\text{relic}} = \Omega_c h^2 \rho_c / s_0 \approx 4.37 \times 10^{-12} \text{ GeV}$.

2. **Bracket:** The search range is $[\text{bracket_factor}[0] \times m_X^{\text{est}}, \text{bracket_factor}[1] \times m_X^{\text{est}}]$ (default factors: 0.1 and 5.0).
3. **Evaluate endpoints:** Run the full Boltzmann solver + background evolution at both bracket endpoints, computing $\log_{10}(m_X Y_X)$.
4. **Linear interpolation:** Guess the solution from linear interpolation in $(\log_{10} m_X, \log_{10}(m_X Y_X))$.
5. **Refine:** Evaluate at $n_{\text{refine}} = 5$ additional points around the guess (spanning a factor of 2 in each direction), and interpolate the final solution.

The background evolution is run with a reference coupling $\varepsilon_{\text{ref}} = 0.1$ (prompt decay, no entropy injection) to isolate the mass dependence. If `find_epsX_min = True`, the function additionally finds the minimum portal coupling for which entropy injection is negligible ($S_{\text{ratio}}(\varepsilon_{\text{min}})/S_{\text{plateau}} - 1 < \delta_S$, with $\delta_S = 0.001$ by default), via root-finding on the entropy ratio.

9 Assumptions, Approximations, and Validity

9.1 Fundamental Assumptions

1. **Thermal equilibrium at early times.** The dark sector is assumed to be in thermal equilibrium with itself (but not necessarily with the SM) at the start of the evolution ($x = x_{\text{min}}$). Both species start with $\bar{\mu} = 0$.
2. **Maxwell-Boltzmann statistics.** The equilibrium distributions, all Boltzmann equation collision terms, and the dark-sector contribution to the Hubble rate use Maxwell-Boltzmann statistics ($f \approx e^{-E/T}$) throughout the solver. This is accurate for $z = m/T \gtrsim 3$ (errors $\lesssim 1\%$) but can introduce $\mathcal{O}(10\%)$ errors on n^{eq} near $z \sim 1$. The approximation enters in several places:
 - *Equilibrium number densities* $n_{\text{MB}}^{\text{eq}}(z)$ and $\lambda_i(z)$, used to set Y_i^{eq} and track departures from equilibrium.
 - *Collision integrals:* the thermal averages $\langle\sigma v\rangle$ and $\langle\sigma v^2\rangle$ rely on the factorisation $f_1 f_2 = e^{-(E_1+E_2)/T}$, which is exact only for MB.
 - *Chemical-potential parametrisation:* the identity $Y_i = Y_i^{\text{eq}} e^{\bar{\mu}_i}$ is an MB result; with full Bose-Einstein or Fermi-Dirac statistics, $n(\mu)$ depends nonlinearly on μ inside the phase-space integral.
 - *Energy moments* ($B_1, B_2, dB_1/dT_h$) and the dark-sector energy density $\rho_{\text{hidden}} = B_1^X n_X + B_1^Y n_Y$ that corrects the Hubble rate.

In practice, the impact on the relic density Ωh^2 is minimal. DM freeze-out occurs at $z_{\text{FO}} \sim 20\text{--}30$, where MB is excellent. The mediator Y (a boson) passes through $z_Y \sim 1$ while still in chemical equilibrium ($\bar{\mu}_Y = 0$), so the $\sim 10\%$ Bose-enhancement error on n_Y^{eq} only mildly shifts the temperature evolution during the equilibrium phase and is washed out well before freeze-out. The dark-sector correction to H_{tot} is itself a sub-leading contribution (the SM radiation dominates), so MB errors there are a second-order effect. Overall, the MB approximation is expected to contribute $\lesssim 1\%$ uncertainty on Ωh^2 across the tested parameter space, well below other systematics.

Note that the SM $g_*(T)$ and $g_{*S}(T)$ are *not* affected: they are loaded from pre-tabulated files computed with full quantum statistics.

3. **Chemical potential additivity.** The yield parametrization $Y_i = Y_i^{\text{eq}} e^{\bar{\mu}_i}$ assumes that chemical potentials are additive and that the departure from equilibrium can be captured by a single number per species. This is valid in the weak-coupling regime ($\bar{\mu}_i \lesssim \text{few}$) but may break down for highly non-thermal distributions.
4. **Kinetic equilibrium within the dark sector.** All dark-sector species are assumed to share a common dark temperature T_h . This requires efficient elastic scattering within the dark sector.
5. **No dark sector–SM kinetic coupling.** The portal coupling ε is assumed small enough that it does not thermalise the dark sector with the SM bath during freeze-out. The dark temperature T_h evolves independently.
6. **Radiation domination.** The SM background is assumed to be radiation-dominated throughout the Boltzmann evolution. Dark-sector energy density is included as a correction to the Hubble rate but does not dominate.

9.2 Cross Section and Decay Width Approximations

1. **Tree-level only.** All dark-sector cross sections and vector-portal decay widths are computed at tree level. No loop corrections, NLO contributions, or resummation effects are included for the dark-sector processes.
2. **Non-relativistic expansion.** The $2 \rightarrow 2$ annihilation cross section is expanded to s -wave + p -wave ($\langle \sigma v \rangle = 2a + 6b/x_X$). Higher partial waves are neglected. In the HiggsPortal, the p -wave contribution is dropped when $x_X < 1$ to avoid numerical issues in the ultra-relativistic regime.
3. **Cannibal rescaling coefficients $\Delta_{1,2}$.** The phenomenological coefficients Δ_1 and Δ_2 (defaulting to 0.5) parametrize the uncertainty in the full $3 \rightarrow 2$ matrix element computation. They serve as $\mathcal{O}(1)$ fudge factors. The exact tree-level Δ_3 (Eq. (19)) is available for the VectorPortal.
4. **No Sommerfeld enhancement or co-annihilation.** The code does not include Sommerfeld enhancement, bound-state effects, or co-annihilation with other dark-sector states. Only $X\bar{X} \rightarrow YY$ annihilation is considered. Sommerfeld enhancement can be significant for $m_Y \ll m_X$ and $\alpha_X \gtrsim 0.01$.
5. **HiggsPortal $\phi XX \rightarrow XX$:** This cannibal channel is set to zero (subdominant, no simplified analytic form).
6. **Vector portal decay widths:** Purely fermionic final states at tree level. No loop-induced $Z' \rightarrow gg, \gamma\gamma$ channels, and no hadronic resonance effects below ~ 2 GeV.
7. **Higgs portal decay width stitching:** The transitions between the dispersive (< 2 GeV), pQCD (2–5 GeV), and HDECAY (> 5 GeV) regimes are not explicitly smoothed. Discontinuities of order a few percent are possible at the boundaries, particularly at $m_\phi \approx 2$ GeV. Above 500 GeV, tree-level widths are used to avoid un-resummed Sudakov logarithms, missing $\mathcal{O}(10\%)$ radiative corrections.
8. **$\phi \rightarrow hh$ coupling:** The singlet-extension coupling $|\lambda_{\phi hh}|^2 / \sin^2 \theta = (2m_\phi^2 + m_h^2)^2 / (4v_H^2)$ is model-dependent. Alternative UV completions may give different results.

9.3 Constraint Approximations

BBN hadronic injection:

1. All light-quark and gluon channels are approximated with the $u\bar{u}$ injection spectrum. Bosonic channels (WW , ZZ , hh) are treated as $u\bar{u}$ -like with the full branching ratio, despite only $\sim 70\%$ of their decays being hadronic (conservative).
2. Outside the tabulated mass range (30 GeV–1000 TeV), the bounds are extrapolated with fixed power-law exponents. For lifetimes exceeding the tabulated range, the bound is clamped to the longest- τ value (conservative).
3. The harmonic combination (Eq. (79)) assumes that the different injection channels contribute independently to nucleosynthesis. Non-linear feedback effects are neglected.

Direct detection:

1. **VectorPortal:** The full Z – Z' mixing formula from arXiv:2509.08043 is used, including isospin violation. Exact at tree level; loop corrections and nuclear structure effects are neglected.
2. **HiggsPortal:** Small-angle approximation $\sin\theta \approx \theta$. Assumes Majorana DM (Dirac would be a factor of 2 smaller). The Higgs-nucleon form factor $f_N = 0.30$ is fixed (lattice QCD gives $f_N \in [0.26, 0.33]$).
3. **LiLjPortal:** Returns $\sigma_{\text{SI}} = 0$ at tree level. Loop-induced contributions through lepton loops could give a non-zero but highly suppressed cross section.

9.4 Solver Limitations

1. **Phase-switching thresholds.** The transitions between solver phases are controlled by numerical thresholds (Γ_{switch} , `cannibal_switch_full`, etc.). Default values work well for the tested parameter space but may need adjustment for extreme parameters.
2. **Convergence detection.** The `dlxixi_NR` convergence criterion is implemented as a terminal `solve_ivp` event, so convergence is detected continuously (not only at grid boundaries). Convergence is only checked after $x > 2x_{\text{FO}}$. For very late freeze-out or large mass ratios, this may terminate the evolution prematurely.
3. **Background solver.** Uses the finite- m_Y s -wave annihilation cross section (Eq. ??) for the DM residual annihilation term. The p -wave contribution is negligible at late times ($x \gg x_{\text{FO}}$) but could matter in edge cases. The optional `correct_Y_decay` flag accounts for mediator decay missed during the Boltzmann phase; this is an instantaneous-injection approximation and may lose accuracy when $\Gamma_Y/H \gg 1$ for a large fraction of the Boltzmann evolution.

10 Parameter Space and Validity Ranges

10.1 Parameter Ranges

Parameter	Tested range	Theoretical validity
m_X (DM mass)	8 GeV–100 TeV	$m_X > m_Y$ required
m_Y (mediator mass)	0.3 GeV–70 TeV	—
$r = m_X/m_Y$	1.3–10	$r > 1$ (kinematic threshold)
α_X (dark coupling)	5×10^{-4} – 10^{-2}	$\alpha_X < 1$ (perturbativity)
ε (portal coupling)	10^{-18} – 10^{-2}	$\varepsilon < 1$
$\lambda_{s,p}$ (Higgs portal)	5×10^{-2} –0.25	$\lambda < 4\pi$
$g_{B-L}, g_B, g_{L_i-L_j}$	1.0 (default)	$g < 4\pi$

The annihilation $X\bar{X} \rightarrow YY$ requires $m_X > m_Y$ (i.e. $r > 1$); the code does not enforce this. For $\alpha_X \gtrsim 0.1$, higher-order corrections become important. The defining assumption $\alpha_X \gg \varepsilon^2$ must hold for the secluded mechanism to operate; if ε is too large, the dark sector thermalises with the SM and the scenario reduces to standard WIMP freeze-out. For $r \gg 1$, the mediator freezes out at $x \sim r^2 \times 10$, requiring $x_{\max} \geq r^2 \times 10^4$; the QSSA solver auto-scales $x_{\max}$ accordingly.

The SM g_* tables cover $T \in [0.1 \text{ MeV}, 100 \text{ GeV}]$ (“`std.tab`”) with Higgs-portal-specific variants (HP_A/B/C). Outside the tabulated range, g_* and g_{*S} are extrapolated with constant edge values.

10.2 Numerical Precision and Regions of Reduced Accuracy

Source	Estimated error
ODE integration (Radau, rtol= 10^{-8})	$\lesssim 0.01\%$ on Y_X
Bessel function asymptotic ($z > 50$)	$\lesssim 10^{-10}$ relative
MB statistics vs. full integral ($z \sim 1$ –3)	1–10% on n^{eq} ; $\lesssim 1\%$ on Ωh^2
g_* interpolation	$\lesssim 1\%$ (smooth tables)
HDECAY decay widths	$\lesssim 5\%$ (NNLO QCD)
BBN bound interpolation	$\lesssim 10\%$ (grid density dependent)
$\Delta_{1,2}$ cannibal coefficients	$\mathcal{O}(1)$ uncertainty

Precision degrades in the following regimes:

- **$m_Y < 2 \text{ GeV}$:** Decay widths rely on dispersive calculations with form-factor uncertainties; the transition region 1–2 GeV is particularly uncertain. For $m_Y \sim 2$ –5 GeV, the pQCD regime is only approximately valid due to non-perturbative QCD effects and resonances. Above 500 GeV, tree-level widths lack NLO corrections ($\mathcal{O}(10\%)$ systematic).
- **Small α_X ($\lesssim 10^{-4}$):** Freeze-out occurs very late ($x_{\text{FO}} \gg 100$), and the solver may require increased $x_{\max}$ and finer grids.
- **Large r ($\gtrsim 10$):** The mediator becomes non-relativistic very late; cannibal processes can maintain Y in equilibrium for a long time, requiring careful choice of $x_{\max}$ and phase-switching thresholds.
- **$m_X \lesssim 10 \text{ GeV}$:** Direct detection limits (LZ) are only valid for $m_X \geq 9 \text{ GeV}$; below this the function returns NaN.
- **Near resonance $m_Y \approx M_Z$:** The VectorPortal couplings (Sec. 3.1) have an unregularised pole at $m_{Z'} = M_Z$ in the denominator ($M_Z^2 - m_{Z'}^2$).
- **$r \rightarrow 1$ limit:** The phase space for $X\bar{X} \rightarrow YY$ closes and the cross section vanishes as $(1 - r_v^2)^{3/2}$; the solver may fail to find freeze-out.

11 Physical Constants

Symbol	Value	Description
<i>Fundamental constants</i>		
M_{Pl}	$2.435 \times 10^{18} \text{ GeV}$	Reduced Planck mass
v_H	246.22 GeV	Higgs vacuum expectation value
m_N	0.9389 GeV	Nucleon mass
$\hbar^2 c^2$	$0.3894 \times 10^{-27} \text{ GeV}^2 \text{ cm}^2$	Conversion factor
$\hbar$	$6.582 \times 10^{-25} \text{ GeV} \cdot \text{s}$	Reduced Planck constant
<i>Electroweak parameters</i>		
$\sin^2 \theta_W$	0.23121	Weinberg angle
α_{EM}	1/137.0	Fine-structure constant
e	0.3029	EM coupling
g_W	0.6520	Weak coupling
g_Y^{SM}	0.3577	Hypercharge coupling
α_s	0.1179	Strong coupling at M_Z
<i>Boson masses</i>		
m_h	125.10 GeV	Higgs mass
M_Z	91.1876 GeV	Z boson mass
M_W	80.379 GeV	W boson mass
Γ_Z	2.4952 GeV	Z total width
<i>Fermion masses</i>		
m_e	$0.511 \times 10^{-3} \text{ GeV}$	Electron
m_μ	0.1057 GeV	Muon
m_τ	1.777 GeV	Tau
m_u	$2.16 \times 10^{-3} \text{ GeV}$	Up quark
m_d	$4.67 \times 10^{-3} \text{ GeV}$	Down quark
m_s	$93.4 \times 10^{-3} \text{ GeV}$	Strange quark
m_c	1.27 GeV	Charm quark
m_b	4.18 GeV	Bottom quark
m_t	172.69 GeV	Top quark
<i>Cosmological observables</i>		
s_0	2891 cm^{-3}	Entropy density today
ρ_c	$1.054 \times 10^{-5} \text{ GeV } h^2 \text{ cm}^{-3}$	Critical density
$\Omega_c h^2$	0.12	Observed DM relic density
m_Y^{relic}	$\Omega_c h^2 \rho_c / s_0$	Reference relic yield $\times$ mass

12 SM Fermion Charge Tables

Hypercharge table (VectorPortal):

Fermion	Mass [GeV]	Q	Y_L	Y_R	N_c
e	5.11×10^{-4}	-1	-1/2	-1	1
μ	0.1057	-1	-1/2	-1	1
τ	1.777	-1	-1/2	-1	1
ν_e	0	0	-1/2	0	1
ν_μ	0	0	-1/2	0	1
ν_τ	0	0	-1/2	0	1
u	2.16×10^{-3}	+2/3	+1/6	+2/3	3
c	1.27	+2/3	+1/6	+2/3	3
t	172.69	+2/3	+1/6	+2/3	3
d	4.67×10^{-3}	-1/3	+1/6	-1/3	3
s	93.4×10^{-3}	-1/3	+1/6	-1/3	3
b	4.18	-1/3	+1/6	-1/3	3

Other portal charges: For $B-L$: quarks carry $(B-L) = +1/3$, leptons carry $(B-L) = -1$. For Baryon: quarks carry $B = +1/3$, leptons are uncharged. For $L_i - L_j$: only the two lepton families involved carry charges ± 1 (quarks are uncharged):

Flavour	+1	-1
$e-\mu$	e, ν_e	μ, ν_μ
$\mu-\tau$	μ, ν_μ	τ, ν_τ
$e-\tau$	e, ν_e	τ, ν_τ

13 Code Architecture

13.1 Module Structure

```

source/
  __init__.py      Public API exports
  constants.py     Physical constants, fermion tables
  cosmology.py     Cosmology class (g*, Hubble, thermo)
  model.py         DarkSectorModel ABC + 5 portals
  solver.py        BoltzmannSolver + find_mXstar
  bbn.py           BBN hadronic injection bounds
  direct_detection.py LZ and neutrino floor interpolators
  SecludedDM.py    Backward-compatible re-exports

phi_decay/
  __init__.py      ensure_hdecay_ready, public API
  phi_decay_hybrid.py Three-regime stitching
  hdecay_interface.py HDECAY Fortran wrapper
  scalar_portal/   Low-mass regime (< 5 GeV)
    api/model.py   LightScalar / HeavyScalar
    decay/         Decay channel implementations
    data/          Form factors, QCD, constants

data/
  direct_detection_data/ LZ, neutrino floor tables
  bbn_constraints_decay/ 12 BBN data files

_legacy/

```

DM_Relic_old.py

Deprecated monolithic version

13.2 Design Principles

- **Stateless models:** Model instances (`VectorPortal`, etc.) carry only their physical parameters and provide rate methods. They hold no reference to a `Cosmology` instance.
- **Solver coupling:** The `BoltzmannSolver` combines a `Cosmology` and a `DarkSectorModel` instance, reading all model-specific physics through the abstract interface.
- **Factory pattern:** The `find_mXstar` function takes a `model_factory` callable (e.g. `lambda mX, mY: VectorPortal(mX=mX, mY=mY, ...)`) that creates model instances for different mass points.
- **Lazy initialisation:** HDECAY compilation and BBN data loading are deferred until first use (cached after that).

References

- [1] A. Berlin, D. Hooper, and G. Krnjaic, “Thermal Dark Matter From A Highly Decoupled Sector,” *Phys. Rev. D*, vol. 94, no. 9, p. 095019, 2016.
- [2] Y. Hu, C. Cesarotti, and T. R. Slatyer, “Testing Viability of Benchmark Dark Matter Models for the Galactic Center Excess,” 9 2025.
- [3] D. Hooper, R. K. Leane, Y.-D. Tsai, S. Wegsman, and S. J. Witte, “A systematic study of hidden sector dark matter: application to the gamma-ray and antiproton excesses,” *JHEP*, vol. 07, no. 07, p. 163, 2020.
- [4] M. W. Winkler, “Decay and detection of a light scalar boson mixing with the Higgs boson,” *Phys. Rev. D*, vol. 99, no. 1, p. 015018, 2019.
- [5] A. Djouadi, J. Kalinowski, and M. Spira, “HDECAY: A Program for Higgs boson decays in the standard model and its supersymmetric extension,” *Comput. Phys. Commun.*, vol. 108, pp. 56–74, 1998.
- [6] A. Djouadi, J. Kalinowski, M. Muehlleitner, and M. Spira, “HDECAY: Twenty₊₊ years after,” *Comput. Phys. Commun.*, vol. 238, pp. 214–231, 2019.
- [7] M. Kawasaki, K. Kohri, T. Moroi, and Y. Takaesu, “Revisiting Big-Bang Nucleosynthesis Constraints on Long-Lived Decaying Particles,” *Phys. Rev. D*, vol. 97, no. 2, p. 023502, 2018.